Implementing a Compiler

Kenjiro Taura

Contents

The MinC (“Minimum C”) language

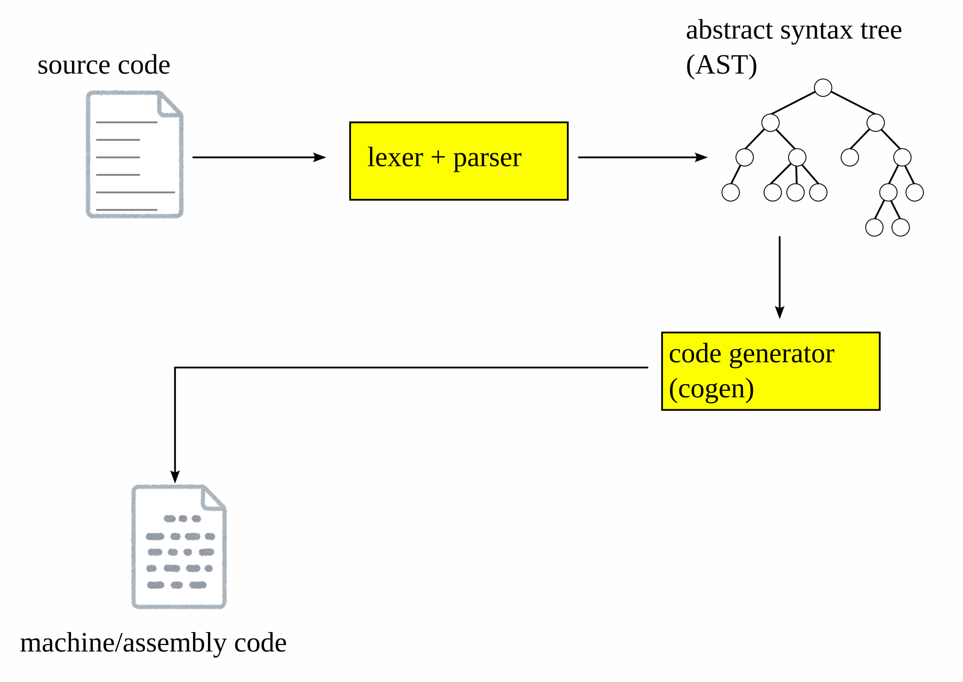
MinC ("Minimum C") spec overview

- all expressions have type `long` (64 bit integer)
  - no other integers, floating point numbers, pointers, or structs
  - everything is long $\Rightarrow$ *type checks are unnecessary*
- no global variables or `typedef`
  - $\Rightarrow$ a program = list of *function definitions*
- supported complex statements are `if`, `while`, and compound statement (`{ ... }`) only
- function calls follow the C convention $\Rightarrow$ MinC code can call or be called by functions compiled by other compilers (e.g., gcc)

Overview of Inside a Compiler

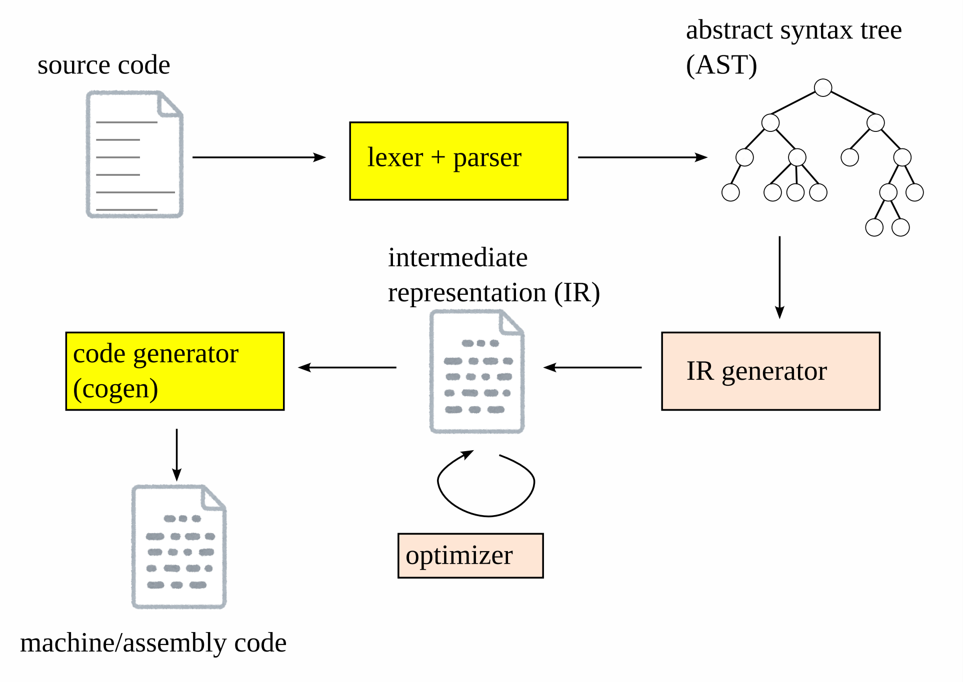
Data structures

- **Abstract Syntax Tree (AST):** data structure representing the program



Data structures

- **Abstract Syntax Tree (AST):** data structure representing the program
- **Intermediate Representation (IR):** common representation portable across multiple source/ target languages



Typical compilation steps

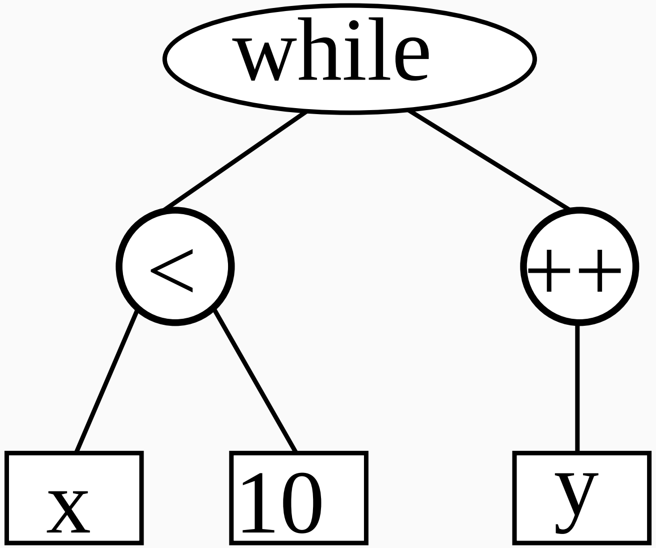
1. **lexing and parsing:** source code (string) $\rightarrow$ AST
2. IR generation: AST $\rightarrow$ IR $(*)$
3. optimization: IR $\rightarrow$ IR $(*)$
4. **code generation:** IR $\rightarrow$ assembly

$(*)$: optional steps

Abstract Syntax Tree (AST)

- a data structure that naturally represents a program

- expression,
- statement,
- function definition,
- the whole program,
- ...



- also called **parse tree**

Components of the baseline code

- `{go,jl,ml,rs}/minc/`

`lex.??`	lexer or tokenizer (string $\rightarrow$ stream of tokens)
`ast.??`	abstract syntax tree (AST) definition
`parse.??`	parser (stream of tokens $\rightarrow$ AST)
`codegen.??`	AST $\rightarrow$ assembly
`main.??` or `minc.??`	main driver

Your work

1. read given, understand, and explain how they work
2. {`lex`,`ast`,`parse`} all lack code for `while` statement
   - ⇒ add necessary definitions for `while` statement
3. implement `codegen`, which is almost empty

Lexer and parser : source code → AST

Lexer and parser

- **lexer:** string $\rightarrow$ sequence of ***tokens*** ($\approx$ words)
  - also called ***lexical analyzer***, or ***tokenizer***
  - `while (x < 10) y++;` $\Rightarrow$

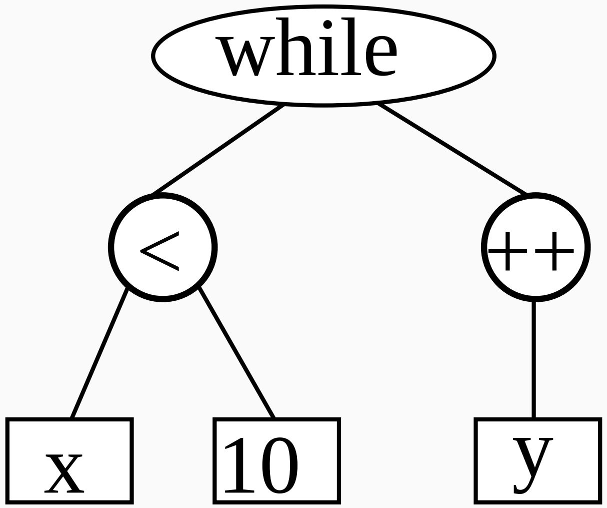
while	(	x	<	10	)	y	++	;
`WHILE`	`LP`	`ID`	`CMP`	`INT`	`RP`	`ID`	`PLSPLS`	`SEMICOLON`

Lexer and parser

- **parser:** sequence of tokens → AST

while	(	x	<	10	)	y	++	;
WHILE	LP	ID	CMP	INT	RP	ID	PLSPLS	SEMICOLON

⇒



Specifying a grammar

- a grammar for ***tokens***
  - specifies which character sequence constitutes a valid token
  - typically uses ***Regular Expressions (RE)***
- a grammar for ***the entire program***
  - specifies which token sequence constitutes a valid input
  - typically uses (a subset of) ***Context Free Grammar (CFG)***
- note: there is an approach that uses a single grammar for both

Regular expression

- a regular expression is any expression that can be formed by:

$$\begin{array}{ll} \varepsilon & \text{(empty string)} \\ c & \text{(a character)} \\ E\ E & \text{(concatenation)} \\ E \mid E & \text{(alternation)} \\ E* & \text{(zero or more repetition)} \\ (E) & \text{(paren)} \end{array}$$

where E is a regular expression

- |, *, (and) are literals

Regular expression

- expressions for convenience

$$
\begin{aligned}
E+ &\equiv E\,E* && \text{(one or more repetition)} \\
E? &\equiv \varepsilon \mid E && \text{(optional)}
\end{aligned}
$$

Regular expression examples

- to build complex expressions, use symbols to represent regular expressions used in other regular expressions. e.g.,

nz	=	1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9	1, 2, …, 9
digit	=	0 | nz	0, 1, 2, …, 9
non_neg	=	0 | nz digit*	0, 12, 34
int	=	-? non_neg	0, −0, 12, −34
fraction	=	int (. digit*)?	−12.34
float	=	fraction (e int)	−12.34e-5
alpha	=	A | B | … Z | a | b | … z	A, B, …, Z, a, b, …, z
alpha_	=	alpha | _	A, B, …, Z, a, b, …, z, _
id	=	alpha_ (alpha_ | digit)*	a, abc, a0_b1

Regular expression semantics (just for formality …)

- a regular expression E represents *a set of strings*, written $[\![E]\!]$

$$
\begin{aligned}
[\![\varepsilon]\!] &= \{ \text{“”} \} \\
[\![c]\!] &= \{ c \} \\
[\![E_0\ E_1]\!] &= \{ e_0 + e_1 \mid e_0 \in [\![E_0]\!], e_1 \in [\![E_1]\!] \} \\
[\![E_0 \mid E_1]\!] &= [\![E_0]\!] \cup [\![E_1]\!] \\
[\![E*]\!] &= \{ \text{“”} \} \cup \{ e_0 + e_1 \mid e_0 \in [\![E]\!], e_1 \in [\![E*]\!] \} \\
[\![(E)]\!] &= [\![E]\!]
\end{aligned}
$$

- note: “+” represents string concatenation

Context Free Grammar (CFG)

- specified by a collection of ***production rules***
- a production rule looks like

$$L \to R_0 \ R_1 \ \dots$$

where

- L : a symbol ***(non-terminal)***
- R_i is either
  - a symbol defined by a production rule(s), or
  - a token name (a ***terminal*** symbol)

An example : expressions

expr	$\rightarrow$	int	12, 345, …
expr	$\rightarrow$	id	f, x, i, is_prime, …
expr	$\rightarrow$	unop expr	-x, exp, !a_greater_than_b
expr	$\rightarrow$	expr binop expr	x + y, a * x + b * y + 1, a & b, …
expr	$\rightarrow$	(expr)	3 * (a + 1)
expr	$\rightarrow$	funcall	

- blue symbols (int, id, unop, binop, (,)) are terminals (tokens)
- above rules overlook the fact that some operators (i.e., + and -) can be used as a unary operator and a binary operator

An example : function call

$$
\begin{array}{lll}
\text{funcall} & \rightarrow & \text{id (comma\_exprs)} \\
\text{comma\_exprs} & \rightarrow & \\
\text{comma\_exprs} & \rightarrow & \text{expr} \\
\text{comma\_exprs} & \rightarrow & \text{expr comma\_expr\_star} \\
\text{comma\_expr\_star} & \rightarrow & \\
\text{comma\_expr\_star} & \rightarrow & \text{, expr comma\_expr\_star}
\end{array}
$$

f(x, 2 * y, 1)

An example : statements

$$
\begin{aligned}
\text{stmt} &\rightarrow \text{;} \\
\text{stmt} &\rightarrow \text{continue ;} \\
\text{stmt} &\rightarrow \text{break ;} \\
\text{stmt} &\rightarrow \text{return ;} \\
\text{stmt} &\rightarrow \text{\{ decl* stmt* \}} \\
\text{stmt} &\rightarrow \text{if (expr) stmt (else stmt)?} \\
\text{stmt} &\rightarrow \text{while (expr) stmt} \\
\text{stmt} &\rightarrow \text{expr ;}
\end{aligned}
$$

Notes

- as you have seen,
  - the same symbol L can appear multiple times in the lefthand side (i.e., *alternation*)
  - R_i can be L or any symbol defined earlier or later (i.e., definitions can be *recursive*)

A few shorthands

- we often use shorthands (|, ?, *, +) that have similar meanings with those for RE
- they can be mechanically eliminated
- the above example using the shorthands:

$$
\begin{array}{lcl}
\text{expr} & \rightarrow & \text{int} \mid \text{id} \mid \text{unop expr} \mid \text{expr binop expr} \mid \text{funcall} \\
\text{funcall} & \rightarrow & \text{id (comma\_exprs)} \\
\text{comma\_exprs} & \rightarrow & \mid \text{expr (, expr)}^{*}
\end{array}
$$

CFG semantics (for formality)

- each symbol L represents a set of token sequences ($[\![L]\!]$)
- $[\![L]\!]$ is the set of token sequences that can result by, starting from L, repeatedly replacing a non-terminal symbol to the righthand side of its production rule, until it becomes a sequence of tokens (terminals)

$$
\begin{array}{rcl}
\text{expr} & \rightarrow & \text{funcall} \\
 & \rightarrow & \text{id (comma\_exprs)} \\
 & \rightarrow & \text{id (expr comma\_expr\_star)} \\
 & \rightarrow & \text{id (id comma\_expr\_star)} \\
 & \rightarrow & \text{id (id , expr comma\_expr\_star)} \\
 & \rightarrow & \text{id (id , expr + expr comma\_expr\_star)} \\
 & \rightarrow & \text{id (id , id + expr comma\_expr\_star)}
\end{array}
$$

CFG semantics (for formality)

$$\begin{aligned} &\rightarrow \; \text{id (id , id + int comma\_expr\_star)} \\ &\rightarrow \; \text{id (id , id + int)} \end{aligned}$$

$$\therefore \text{id (id , id + int)} \in [\![\text{expr}]\!]$$

$$\text{e.g., f (x, y + 1)} \in [\![\text{expr}]\!]$$

An alternative semantics

- $[\![.]\!]$ is the minimal set of token sequences satisfying:

1. $[\![\, t \,]\!] = \{\, t \,\}$ (t : terminal)
2. $L \to R_0 \ldots R_{n-1}$ implies

$$r_0 \in [\![R_0]\!], \ldots, r_{n-1} \in [\![R_{n-1}]\!]$$
$$\Rightarrow r_0 + \ldots + r_{n-1} \in [\![L]\!]$$

- “+” represents concatenation of token sequences

CFG is more expressive than RE

- as you might have noticed, RE is a special case of CFG
- all the constructs of RE can be straightforwardly expressed with CFG, where a token = a character
- e.g., a CFG equivalent to RE "int = 0 | nz digit*"

$$
\begin{aligned}
\text{int} &\to 0 & \text{digit} &\to 0 \\
\text{int} &\to \text{nz digits} & \text{digit} &\to \text{nz} \\
\text{digits} &\to & \text{nz} &\to 1 \mid \ldots \mid 9 \\
\text{digits} &\to \text{digit digits}
\end{aligned}
$$

In general …

- below, CFG(e, L) is a function that converts regular expression e to an equivalent CFG s.t., $[\![L]\!] = [\![e]\!]$

$$
\begin{aligned}
\mathrm{CFG}(\varepsilon, L) &= \{L \rightarrow\} \\
\mathrm{CFG}(c, L) &= \{L \rightarrow c\} \\
\mathrm{CFG}(E_0\ E_1, L) &= \{L \rightarrow R_0\ R_1\} \cup \mathrm{CFG}(E_0, R_0) \cup \mathrm{CFG}(E_1, R_1) \\
\mathrm{CFG}(E_0 \mid E_1, L) &= \{L \rightarrow R_0, L \rightarrow R_1\} \cup \mathrm{CFG}(E_0, R_0) \cup \mathrm{CFG}(E_1, R_1) \\
\mathrm{CFG}(E^*, L) &= \{L \rightarrow \mid R\ L\} \cup \mathrm{CFG}(E, R) \\
\mathrm{CFG}((E))] &= \mathrm{CFG}(E, L)
\end{aligned}
$$

- R, R_0 and R_1 are unique symbols that do not appear elsewhere

A CFG that cannot be expressed by RE

- intuitively, RE *can repeat (E*) but cannot recurse*
- e.g., both "$A \to \mid a\ A$" and "$A \to \mid A\ a$" *can* be expressed by an RE (both are equivalent to a^*), but

$$A \to \mid a\ A\ b$$

cannot ($[\![A]\!] = \{\varepsilon, ab, aabb, aaabbb, \ldots\} = \{a^n b^n \mid n \geq 0\}$)

- the proof is interesting but omitted

If RE $\subset$ CFG, why use both (not just CFG)?

- parsing *general* CFG is expensive ($O(\text{length}^3)$)
- the primary reason is handling *alternatives* requires *backtrack*

$$A \to B_0\ B_1\ \ldots \mid C_0\ C_1\ \ldots \mid D_0\ D_1\ \ldots$$

- practical parsers take either of the following two approaches
  1. allow only alternatives that can be determined with a ***limited lookahead (LL(1), LALR(1), etc.)***
  2. allow backtrack with programmer-supplied *cut points* (*Parsing Expression Grammar; PEG*)

CFG with a limited lookahead (LL(1), LALR(1), etc.)

- recall the syntax of statement

$$
\begin{aligned}
\text{stmt} \;\rightarrow\; & \;;\;|\;\text{continue}\;;\;|\;\text{break}\;;\;|\;\text{return}\;;\;|\;\{\;\text{decl* stmt*}\;\} \\
& |\;\text{if}\;(\;\text{expr}\;)\;\text{stmt}\;(\;\text{else stmt}\;)?\;|\;\text{while}\;(\;\text{expr}\;)\;\text{stmt} \\
& |\;\text{expr}\;;
\end{aligned}
$$

- upon parsing a statement, which branch we should take can be determined just by its *first* token
- *it is essential to have a separate tokenizer for this type of grammar* (looking ahead a token $\neq$ looking ahead a character)

How to Write A CFG Parser in Practice

1. use parser-generator tool
   - yacc/bison (C/C++)
   - Menhir (OCaml)
   - ANTLR4
   - JavaCC
   - …
2. manually write LL(1) parser

- This course chooses LL(1) for lack of a parser generator common across the four languages
- The experience helps you write a robust parser in other occasions

How to Write LL(1) Parser

- For each symbol A (terminal or non-terminal), define a function,

parse_A(tokens),

which takes a stream of tokens and consume its prefix making A

- *tokens* supports the following interface
  - **peek(tokens)** : the "current" (the first unconsumed) token
  - **advance(tokens)** : consume the current token

How to Write LL(1) Parser

1. If A is a terminal (a token)

```
parse_A(tokens) {
  if (peek(tokens) != A) parse_error();
  advance(tokens);
}
```

2. If A is a non-terminal and its definition has $A \to B\ C\ D$,

```
parse_A(tokens) {
  parse_B(tokens);
  parse_C(tokens);
  parse_D(tokens);
}
```

How to Handle Alternatives

3. If A has alternatives, e.g.,
   - $A \to B\ C$
   - $A \to D\ E$

we choose the appropriate branch by "peeking the current token"

```
parse_A(tokens) {
  if (peek(tokens) in FIRST(B C)) {
    parse_B(tokens);
    parse_C(tokens);
  } else if (peek(tokens) in FIRST(D E)) {
    parse_D(tokens);
    parse_E(tokens);
  } else { parser_error(); } }
```

- `FIRST( • )` is the set of tokens that could start ' • '
- The code assumes `FIRST(B C)` and `FIRST(D E)` do not overlap (LL(1))
- You can compute `FIRST( · )` by an algorithm, but for simple cases manual inspection suffices

Limitation of LL(1)

- It has trouble with “left recursion”
- e.g.,
  - $A \to$
  - $A \to A\ b$
- writing parse_A following the above formula would result in infinite recursion
- in practice, we get rid of left recursion by manual inspection
- e.g., the above is equivalent to b*, which is
  - $A \to$
  - $A \to b\ A$

Code generation

Code generation (cogen) — basic structure

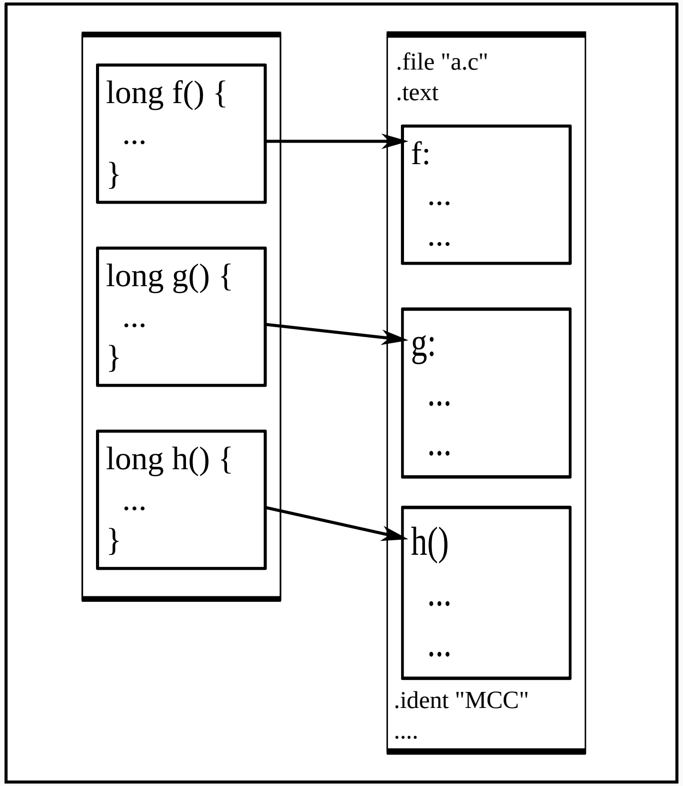
- takes an AST and returns machine code (a list of instructions)
- generate machine code for an AST $\approx$ generate machine code of its components and properly arrange them
- program $\rightarrow$ function definition $\rightarrow$ statement $\rightarrow$ expression

Code generation (cogen) — basic structure

- code generator has lots of:
  - case analysis based on the type of the tree; use:
    - pattern matching (match à la OCaml and Rust), or
    - polymorphism (OCaml objects, Julia function, Go interface, Rust trait)
  - recursive calls to child trees

Compiling an entire file

- ≈ concatenate compilation of individual function definitions

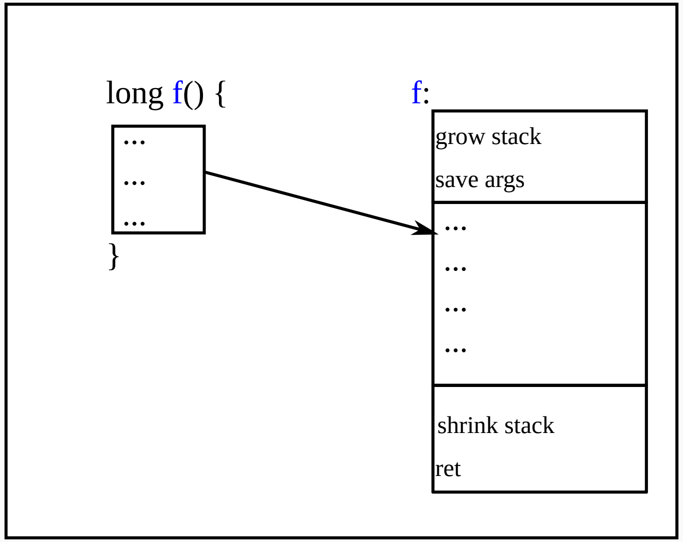


Pseudo code:

```
gen_program (Program([d0, d1, ...])) ... =
    ...
    header
  + (gen_def d0 ...)
  + (gen_def d1 ...)
  + ...
  + trailer
```

Compiling a function definition

- ≈ prologue (grow the stack, etc.) + code for the body (statement) + epilogue (shrink the stack, `ret`, etc.)

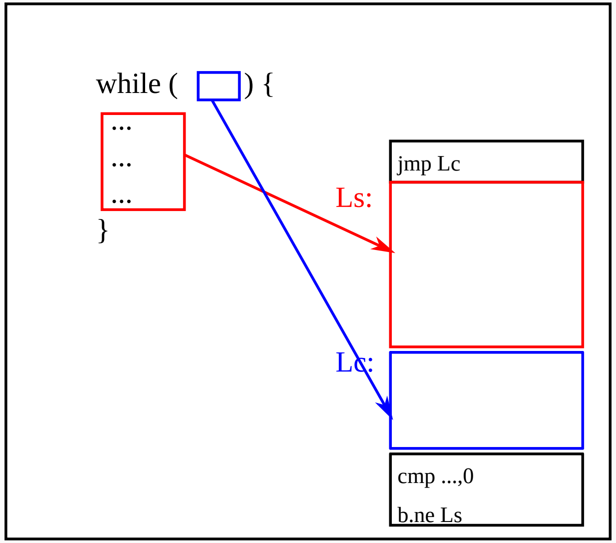


Pseudo code:

```
gen_def (DefFun(f, params, ret_type, body)) =
    (gen_prologue f ...)
  + (gen_stmt body ...)
  + (gen_epilogue f ...)
```

Compiling a statement (while statement)

- ≈ jump to the condition expression; body; the condition expression; compare and conditional branch

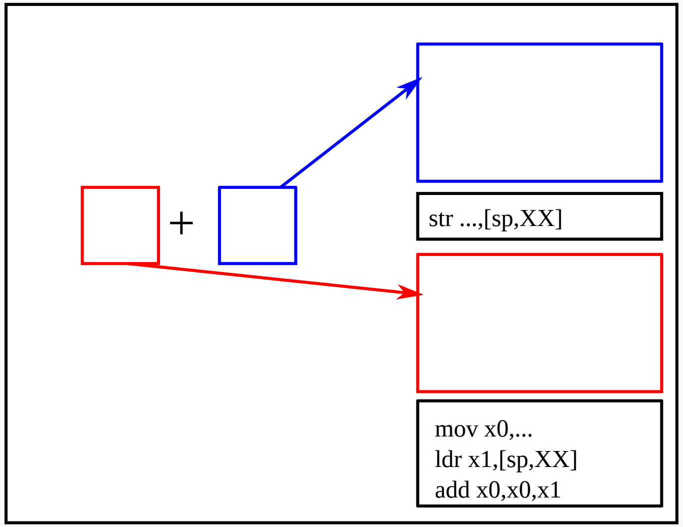



```
gen_while_stmt (StmtWhile(cond, body)) ... =
   cond_op,cond_insns = gen_expr cond ... ;
   body_insns = gen_stmt body ... ;
     ...
   [ jmp Lc; Ls ]
 + body_insns
 + [ Lc ]
 + cond_insns
 + [ cmp cond_op,0; jne Ls ]
```

- (`gen_expr` *expr* …) returns a pair: (instructions to evaluate *expr*, the location of the result)

Compiling an expression (arithmetic)

- ≈ instructions to evaluate the arguments; the arithmetic instruction



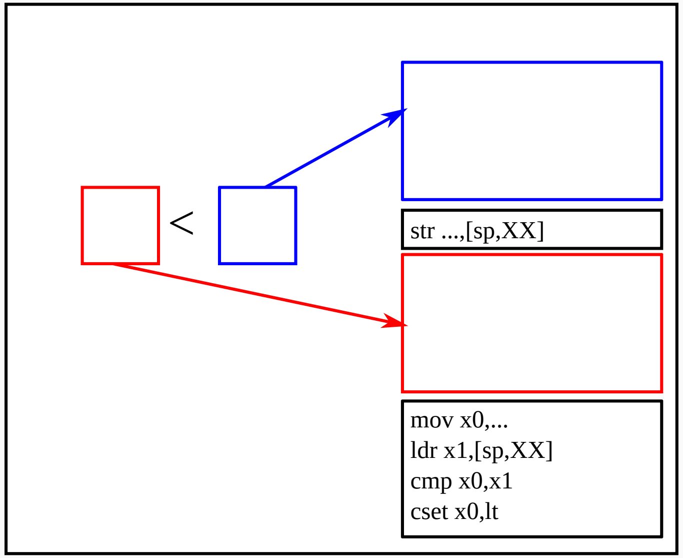

```
gen_add_expr ExprOp("+", [e0; e1]) ... =
  insns1,op1 = gen_expr e1 ... ;
  insns0,op0 = gen_expr e0 ... ;
  m = (* a slot on the stack for e1 *);
  (  insns1
   + [ str op1,m ]
   + insns0
   + [ mov x0,op0;
       ldr x1,m;
       add x0,x0,x1 ],
   x0)
```

Compiling an expression (comparison)

- $A < B$ is an expression that evaluates to:
  - 1 if $A < B$
  - 0 if $A >= B$
- this can be done by cmp + conditional set (cset)

Compiling an expression (comparison)

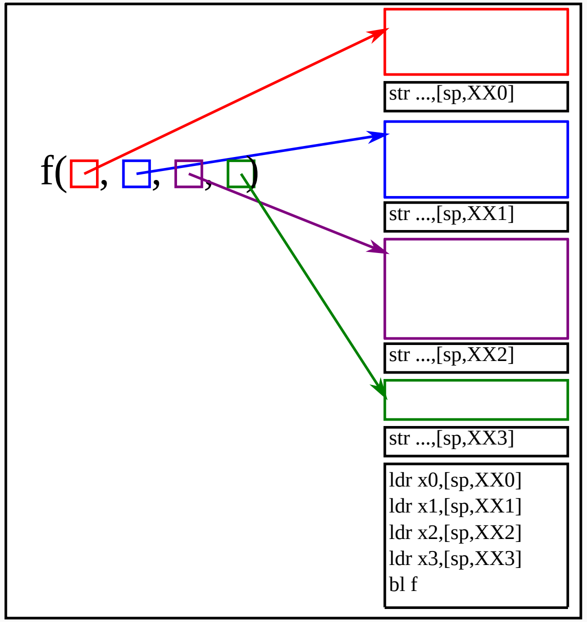
- ≈ compile the arguments; compare; conditional set




```
gen_cmp_expr (ExprOp("<", [e0; e1])) ... =
  insns1,op1 = gen_expr e1 ... ;
  insns0,op0 = gen_expr e0 ... ;
  m1 = (* a slot on the stack for e1 *);
    ...
  (insns1
   + [ str op1,m1 ]
   + insns0
   + [ mov x0,op0;
       ldr x1,m1;
       cmp x0,x1;
       cset x0,lt ],
   x0)
```

Compiling an expression (function call)

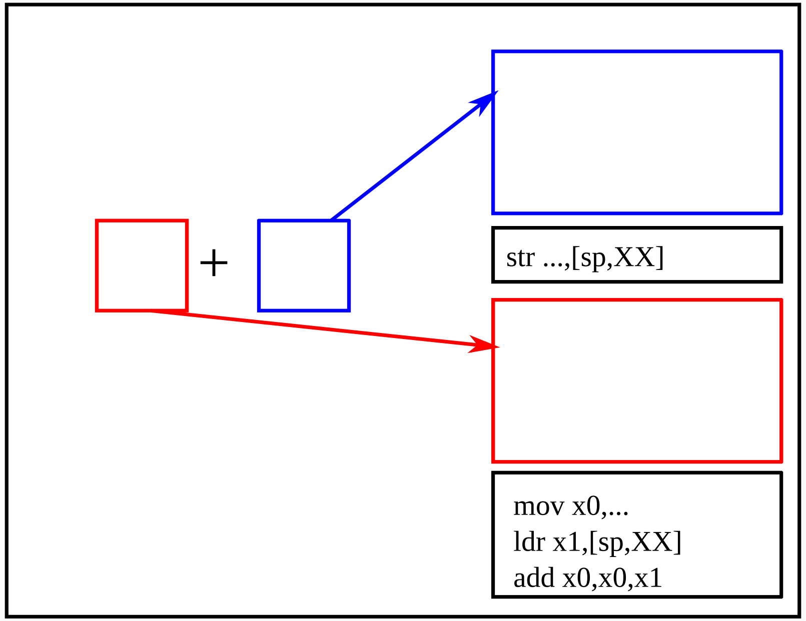
- ≈ instructions for all arguments; put them to positions specified by ABI; a `bl` instruction




```
gen_call_expr (ExprCall(f, [e0;e1;...])) ... =
  [(i0,op0);(i1,op1);..],[m0;m1;...]
    = gen_exprs [e0;e1;...] ...;
  (  (i0 + [str op0,m0])
   + (i1 + [str op1,m1])
   + ...
   + [ldr x0,m0;
      ldr x1,m1;
      ...;
      bl f],
   x0)
```

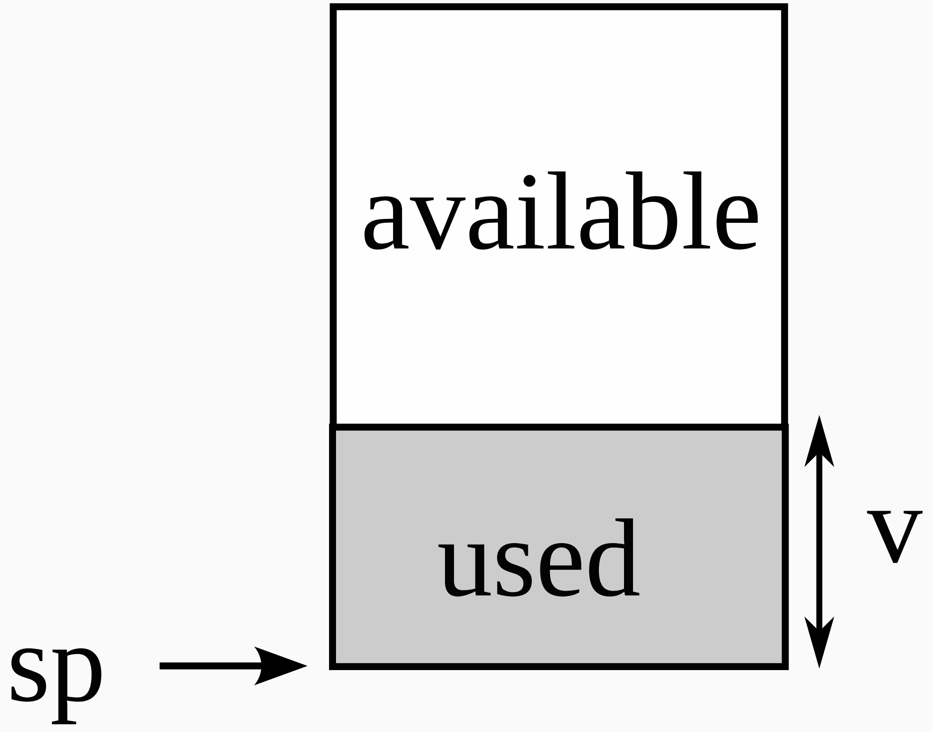
A few left-out details

- how to determine locations to save values of *subexpressions* and *variables*?
- that is, how to determine XX below:



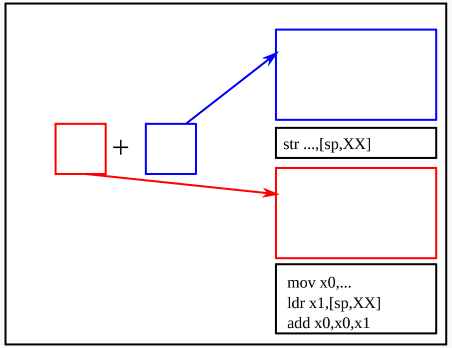
Determining where to save subexpressions

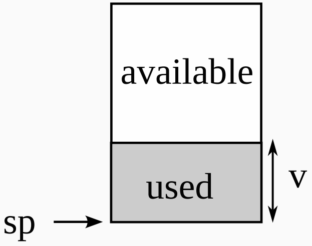
- `gen_expr E` receives a value (`v`) pointing to the lowest end of free space in the current stack frame
- `gen_expr E` v `...` generates instructions that evaluate `E` using (destroying) only addresses at or above SP+v



Determining where to save subexpressions

- when evaluating $A + B$,
  1. evaluate B, using SP+v and higher; save the result at SP+v
  2. evaluate A, using $v + 8$ and higher addresses






```
gen_add_expr ExprOp("+",[e0;e1]) v ... =
  insns1,op1 = gen_expr e1  v    ... ;
  insns0,op0 = gen_expr e0 (v+8) ... ;
  (  insns1
   + [ str op1,[sp,v] ]
   + insns0
   + [ mov x0,op0;
       ldr x1,[sp,v];
       add x0,x0,x1],
   x0)
```

Locations to hold variables

```
if (...) {
  long a, b, c;
  ...
}
```

we obviously need to store `a, b, c` somewhere, but where?

- the problem is almost identical to saving values of subexpressions
- $\rightarrow$ `gen_stmt` also takes v pointing to the free space
- (`gen_stmt` S v ...) generates instructions to execute S, using only addresses at or above $\mathrm{SP} + v$
- $\Rightarrow$
  - $a \mapsto (\mathrm{SP} + v)$
  - $b \mapsto (\mathrm{SP} + v + 8)$
  - $c \mapsto (\mathrm{SP} + v + 16)$

Environment: records where variables are held

- variable locations must be known when generating code for expressions referencing them
  - e.g., to compile `x + 1`, we need to know where `x` is stored
- $\Rightarrow$ make a data structure that holds a mapping: ***variable*** $\mapsto$ ***location*** **(*environment*)** and pass it to `gen_stmt` and `gen_expr`
  - generating code for variables look up the environment
  - a compound statement (`{ ... }`) adds new mappings to the environment

gen_expr receives an environment

```
gen_expr (ExprId(x)) env v =
  m = env_lookup x env;
  ([ ldr x0,m ], x0)
```

`env_lookup x env` searches environment `env` for `x` and returns its location

gen_stmt receives an environment too

```
gen_stmt (StmtCompound(decls, stmts)) env v =
  env', v' = env_extend decls env v;
  gen_stmts stmts env' v' ...
```

- `env_extend decls env v`:
  - assigns locations ($v, v + 8, v + 16, ...$) to variables declared in `decls`
  - registers them in env
  - returns the new environment `env'` and the new free space `v'`

Implementing environment

- an environment is a list of (*variable name*, *location*) pairs (*association list*)
- `v = env_lookup x env`
  - returns the location paired with `x` in environment `env`
- `env' = env_add x v env`
  - returns a new environment `env'` which has a new mapping `x` $\mapsto$ `v` in addition to `env`
- `(env', v') = env_extend decls v env`
  - can be easily built on `env_add` (left for you)

Intermediate Representation (IR)

Intermediate Representation (IR)

- a common representation of programs used by a compiler
- roughly $\approx$ an assembly with unlimited variables
- purposes
  1. achieve portability
     - hopefully independent from the source language (C, C++, Rust, Go, Julia, etc.)
     - hopefully independent from the target language (x86, ARM, PowerPC, etc.)
  2. formulate optimizations as IR $\rightarrow$ IR transformations
- **note:** in the exercise you could design your IR, but it is not necessary (it is possible to directly go from AST $\rightarrow$ asm)

Optimizations performed on IR level (a brief)

- **constant folding and propagation** — compute values at compile time where possible
- **hoisting** — lift instructions in a loop outside of it
- **function call inlining** — replace a call to a function with its body
- **register allocation** — assign registers to variables to reduce memory access